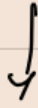


optimization



till now just retrieving data

but now how quickly how efficiently can it
be retrieved



latency reduction.

indexing (B tree).



text index - page (basically labeling the data
book. and dividing it into
smaller chunks).

Clustered index and clustering.

→ physically sorting the
data based on ⁴ primary
key ⁴

Non-Clustered index

→ non-primary key columns

↓
no id column.

↓
but creates reference ids.
in memory
for a non primary
key.

↓
Scans whole table

→ "EXPLAIN"

↓

when we fetch based on non-primary
Key

all the records are scanned to fetch the
data.

→ But when indexed based on non-primary
key and fetched based on a reference.

→ But we need the knowledge of commonly
used non-primary keys for filtering.

→ reduces the fetch times substantially over
large data.

→ Natural Key and surrogate Key

↓
By nature it is
unique identifier.

instead of 1, 2, 3

we can have SSN, address.

↓
System generated
primary
key, AUTO
Increment.

- Use select specific columns.
- Use where before groupby.
- Use joins instead of subquery
- avoid inbuilt functions ~~of~~ on indexed columns.
 - ↓
 - use between.
- Use limit.
- Use CTE instead of subquery.
- Use Explain for query execution plan.
- Maintenance
 - ↓
 - Analyze, optimize (~~run~~ run periodically).
- Avoid large offsets in pagination.
 - limit 1000, 10: ✗
 - payment > 1000 limit 10 ✓✓